Link Path Network Optimization Report

Model Type: **link_path**

Metric	Value
Status	Optimal
Objective Value	$1,650.00
Total Revenue	$1,650.00
Total Expansion Cost	$0.00
Total Installation Cost	$0.00
Total Cost	$0.00
Net Profit	$1,650.00
Accepted Demands	4 / 5
Rejected Demands	1 / 5
Acceptance Rate	80.0%
Nodes	10
Edges	16
Total Initial Capacity	535.00
Total Expansion	0.00
Capacity Increase (%)	0.0%

Insights & Highlights

Key Insight	Detail
Most Congested Edge	R3 -> R5 (100.0%)
Number of Highly Utilized Edges (≥90%)	3
Most Capacity Expansion	None
Total Lost Revenue	$250.00
Number of Expanded Edges	0

WARNING: Congestion risk in the following edges (utilization ≥90%):
- Edge R1 -> R3: 100.0% used
- Edge R2 -> R5: 100.0% used
- Edge R3 -> R5: 100.0% used

Accepted Demands & Flow Paths

Demand 0: S1 -> D1

Bandwidth: 25.0 units **Revenue:** $400.0
Revenue/BW: $16.00/unit

Path	Fraction	Bandwidth
S1 -> R1 -> R3 -> D1	100.00%	25.00

Demand 1: S1 -> D2

Bandwidth: 30.0 units **Revenue:** $450.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S1 -> R2 -> R5 -> D2	100.00%	30.00

Demand 2: S2 -> D2

Bandwidth: 20.0 units **Revenue:** $300.0
Revenue/BW: $15.00/unit

Path	Fraction	Bandwidth
S2 -> R2 -> R4 -> D2	100.00%	20.00

Demand 3: S2 -> D3

Bandwidth: 35.0 units **Revenue:** $500.0
Revenue/BW: $14.29/unit

Path	Fraction	Bandwidth
S2 -> R2 -> R5 -> D3	14.29%	5.00
S2 -> R3 -> R5 -> D3	85.71%	30.00

Rejected Demands

Idx	Source	Target	Bandwidth	Revenue ($)
4	S1	D3	15.0	250.0

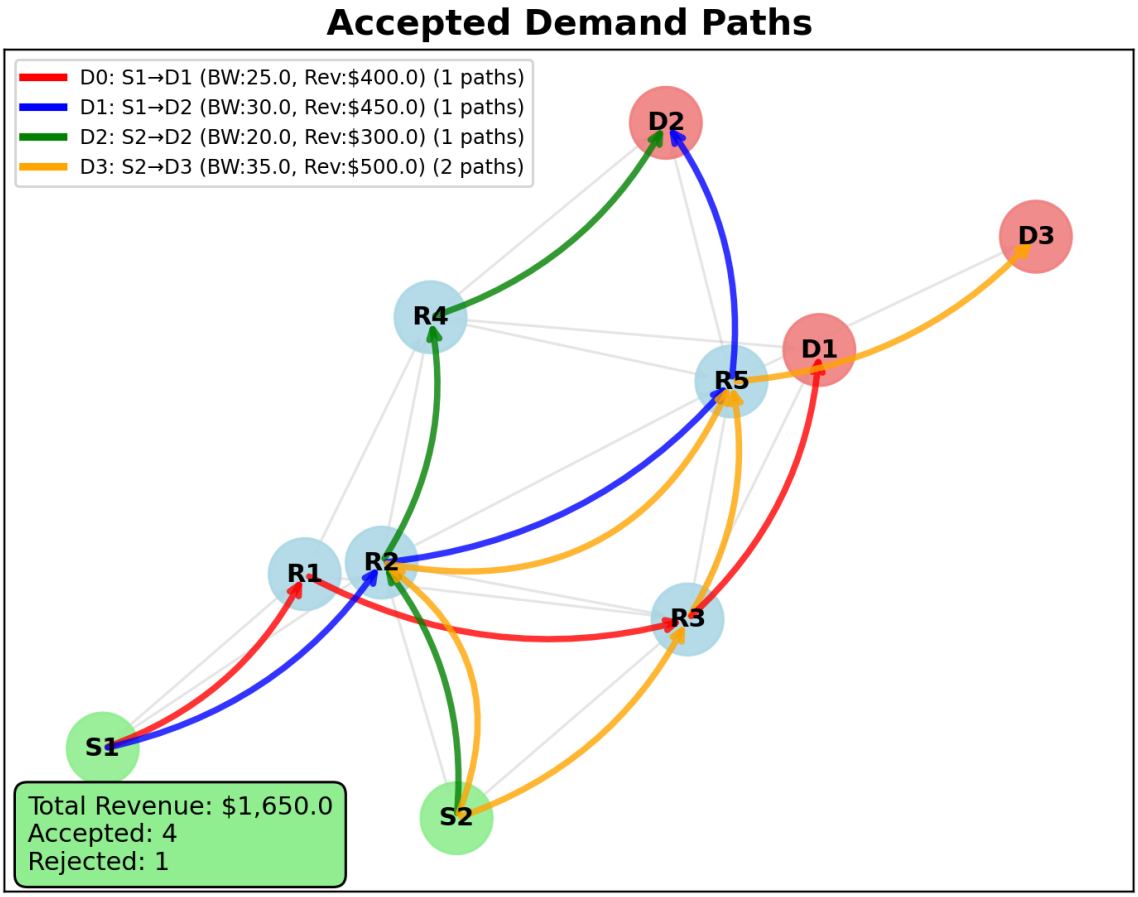
Total Lost Revenue: $250.00

Edge Utilizations (Top 12 Edges)

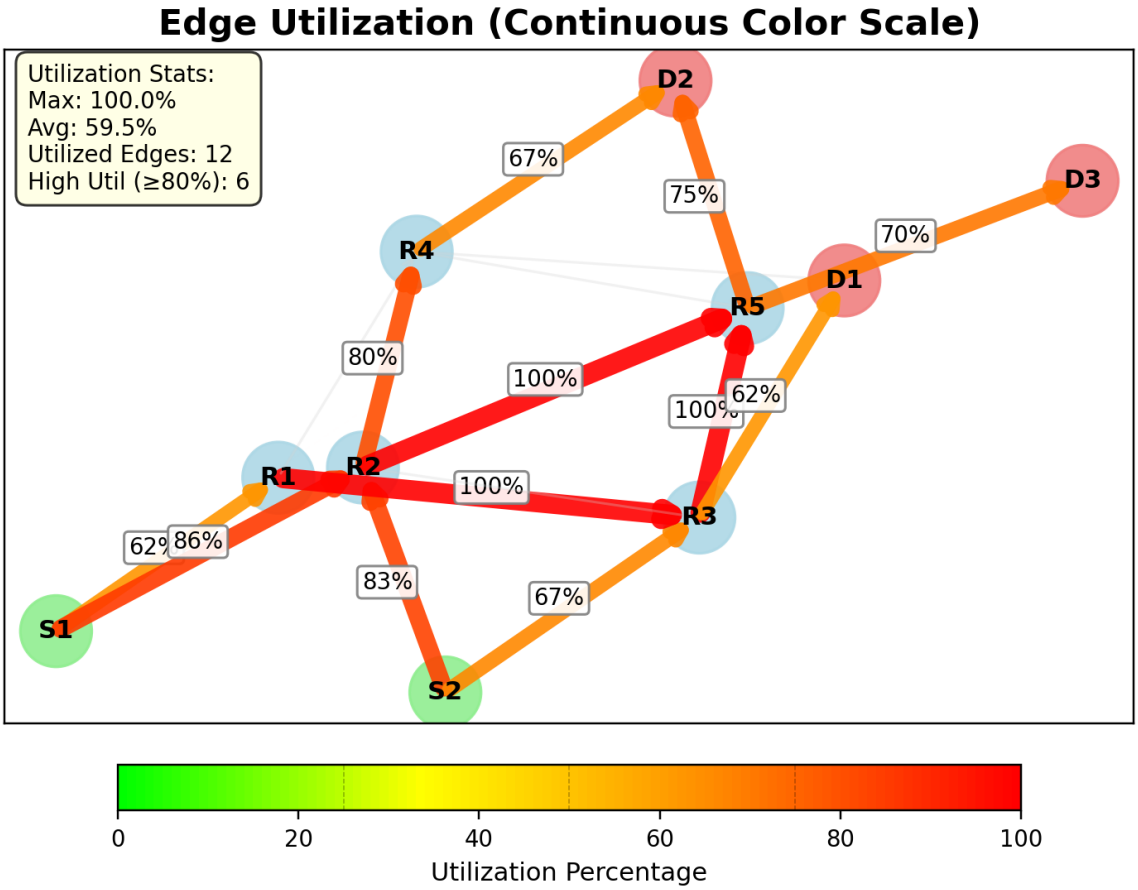
Edge	Flow	Total Capacity	Utilization (%)
R3 -> R5	30.00	30.00	100.0 %
R1 -> R3	25.00	25.00	100.0 %
R2 -> R5	35.00	35.00	100.0 %
S1 -> R2	30.00	35.00	85.7 %
S2 -> R2	25.00	30.00	83.3 %
R2 -> R4	20.00	25.00	80.0 %
R5 -> D2	30.00	40.00	75.0 %
R5 -> D3	35.00	50.00	70.0 %
S2 -> R3	30.00	45.00	66.7 %
R4 -> D2	20.00	30.00	66.7 %
S1 -> R1	25.00	40.00	62.5 %
R3 -> D1	25.00	40.00	62.5 %

Network Analysis Visualizations

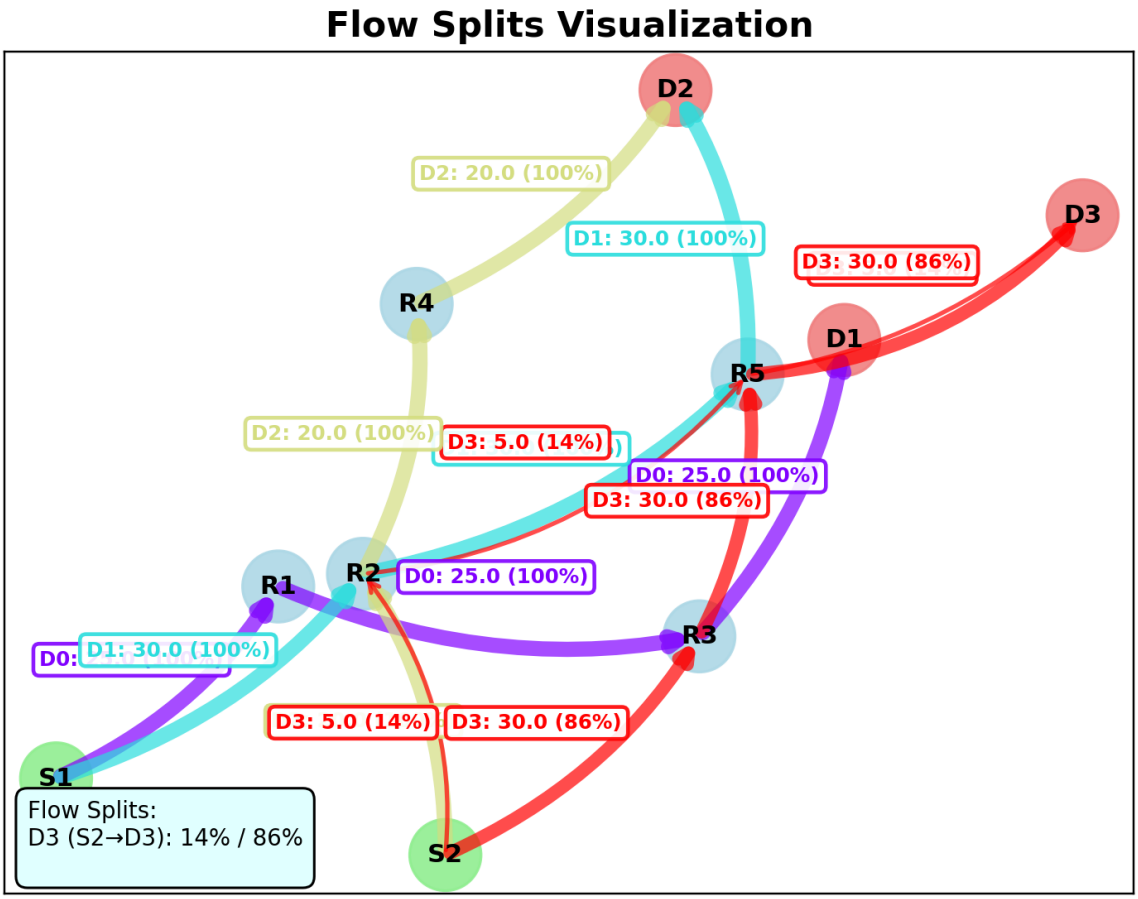
Accepted Demand Paths



Edge Utilization



Flow Splits


Flow Splits Visualization
D2
D3
D1
R4
R5
R2
R1
R3
S1
S2
D2: 20.0 (100%)
D1: 30.0 (100%)
D3: 30.0 (86%)
D2: 20.0 (100%)
D3: 5.0 (14%)
D0: 25.0 (100%)
D3: 30.0 (86%)
D0: 25.0 (100%)
D1: 30.0 (100%)
D3: 5.0 (14%)
D3: 30.0 (86%)
Flow Splits:
D3 (S2→D3): 14% / 86%